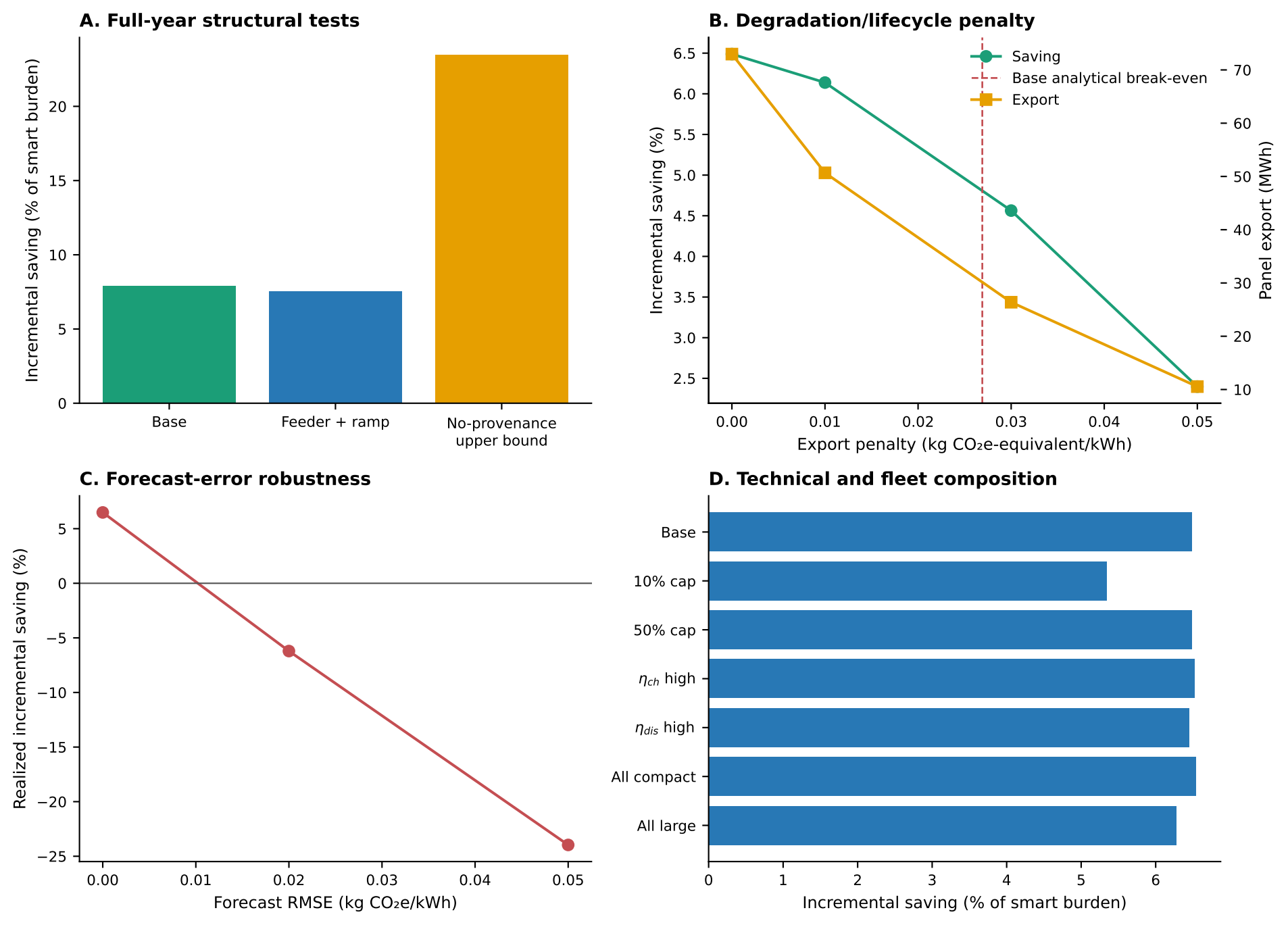
A. Full-year structural tests
Incremental saving (% of smart burden)
0
5
10
15
20
Base
Feeder + ramp
No-provenance upper bound
B. Degradation/lifecycle penalty
Incremental saving (%)
2.5
3.0
3.5
4.0
4.5
5.0
5.5
6.0
6.5
Panel export (MWh)
10
20
30
40
50
60
70
Saving
Base analytical break-even
Export
0.00
0.01
0.02
0.03
0.04
0.05
Export penalty (kg CO₂e-equivalent/kWh)
C. Forecast-error robustness
Realized incremental saving (%)
−25
−20
−15
−10
−5
0
5
0.00
0.01
0.02
0.03
0.04
0.05
Forecast RMSE (kg CO₂e/kWh)
D. Technical and fleet composition
Base
10% cap
50% cap
η_{ch} high
η_{dis} high
All compact
All large
0
1
2
3
4
5
6
Incremental saving (% of smart burden)